

Combined analysis of climate, technological and price changes on future arable farming systems in Europe



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ABSTRACT

In this study, we compare the relative importance of climate change to technological, management, price and policy changes on European arable farming systems. This required linking four models: the SIMPLACE crop growth modelling framework to calculate future yields under climate change for arable crops; the CAPRI model to estimate impacts on global agricultural markets, specifically product prices; the bio-economic farm model FSSIM to calculate the future changes in cropping patterns and farm net income at the farm and regional level; and the environmental model INTEGRATOR to calculate nitrogen (N) uptake and losses to air and water. First, the four linked models were applied to analyse the effect of climate change only or a most likely baseline (i.e. B1) scenario for 2050 as well as for two alternative scenarios with, respectively, strong (i.e. A1-b1) and weak economic growth (B2) for five regions/countries across Europe (i.e. Denmark, Flevoland, Midi Pyrenées, Zachodniopomorski and Andalucia). These analyses were repeated but assuming in addition to climate change impacts, also the effects of changes in technology and management on crop yields, the effects of changes in prices and policies in 2050, and the effects of all factors together. The outcomes show that the effects of climate change to 2050 result in higher farm net incomes in the Northern and Northern–Central EU regions, in practically unchanged farm net incomes in the Central and Central–Southern EU regions, and in much lower farm net incomes in Southern EU regions compared to those in the base year. Climate change in combination with improved technology and farm management and/or with price changes towards 2050 results in a higher to much higher farm net incomes. Increases in farm net income for the B1 and A1-b1 scenarios are moderately stronger than those for the B2 scenario, due to the smaller increases in product prices and/or yields for the B2 scenario. Farm labour demand slightly to moderately increases towards 2050 as related to changes in cropping patterns. Changes in N₂O emissions and N leaching compared to the base year are mainly caused by changes in total N inputs from the applied fertilizers and animal manure, which in turn are influenced by changes in crop yields and cropping patterns, whereas NH₃ emissions are mainly determined by assumed improvements in manure application techniques. N emissions and N leaching strongly increase in Denmark and Zachodniopomorski, slightly decrease to moderately increase in Flevoland and Midi-Pyrenées, and strongly decrease in Andalucia, except for NH₃ emissions which zero to moderately decrease in Flevoland and Denmark.

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1. Introduction

Conditions for farming over Europe are changing, driven by a multitude of interrelated factors such as a higher integration of EU agriculture in global markets, on-going technological developments (Ewert et al., 2006), climate change (Parry et al., 1999, 2004; Porter et al., 2014) and growing global demand for feed, food, fibre and bio-based energy. Understanding the risks, as well as the opportunities for EU farming

evolving from these partially long-term changes (cf. Reidsma et al., 2015; Wolf et al., 2012 for The Netherlands) can support a rational redesign of the Common Agricultural Policy or others policies impacting EU agriculture. For example, climate change will induce more frequent extreme weather events (Field et al., 2012) and the reduction of water availability in many locations. Both may result in lower yields and/or lower yield quality, and require adaptation measures at farm level which can be supported by an appropriate policy environment. Socio-economic conditions will also change over time dependent on drivers such as degree of globalization and will affect the future crop prices and consequently cropping patterns and intensity, and finally the

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sustainability and environmental impacts of future farming systems. While recent studies have examined future European crop yields and production under climate change (Donatelli et al., 2012), they have not considered how changing technology in combination with climate change will impact crop yields, nor were the macro- (EU) and micro- (farm) economic impacts and environmental impacts evaluated. Integrated approaches that are able to model the interactions between future macro- and micro-economic impacts appear to be essential to derive information on impacts of future changes in climate and other drivers on the sustainability and the environmental impacts of future farming systems (van Ittersum et al., 2008).

While, the demand for integrated studies of climate risk impacts on farming systems is clear (Robertson et al., 2013; Ewert et al., *in press*), few examples of such studies exist (for exceptions, see Fischer et al., 2005; Lehtonen et al., 2010; Nelson et al., 2014). In addition to the various conceptual and methodological challenges associated with linking disciplinary models (Ewert et al., 2009, 2011a, 2011b, *in press*), integrated assessment studies are particularly challenging due to their cross-scale nature: climate, policy and price drivers generally occur at large scales, whereas management decisions are typically made at farm and local levels (Patt et al., 2010).

Our aim in this study is to generate policy relevant information about how EU farming is likely to evolve in response to climate, technological and market changes to 2050. Based on important indicators at farm level we have identified problems, threats and opportunities for current and future EU farming systems. We have linked four existing models (i.e. SIMPLACE, CAPRI, FSSIM and INTEGRATOR). The linked models constitute an impact assessment (IA) tool that captures impacts of these future changes on various economic and environment aspects of EU farms. A meeting with various stakeholders from DG Agriculture and Rural development (http://ec.europa.eu/dgs/agriculture/index_en.htm) initiated this Agri-test case, and framed the subsequent study. The resulting aim of the test case was to assess the impacts of future climatic and other (technological, socio-economic and policy) changes on (i) agronomic, (ii) socio-economic, and (iii) environmental indicators of different farming systems in the EU. The main focus was on five regions: Denmark, as a whole; Flevoland, The Netherlands; Midi Pyrenées, France; Zachodniopomorski, Poland and Andalucia, Spain. With these regions, a wide cross-section of the variation in farming and environmental conditions across the EU is covered. These regions range from cool to warm climates in the sequence of Denmark, Zachodniopomorski, Flevoland, Midi Pyrenées and Andalucia and from extensive (with rather long rotations and low input levels) to intensive arable cropping systems in the sequence of Zachodniopomorski, Andalucia, Denmark, Midi Pyrenées and Flevoland.

2. Methodology

2.1. Model descriptions

SIMPLACE (Scientific Impact assessment and Modelling Platform for Advanced Crop and Ecosystem management) is a modelling framework in which various dynamic and process-based crop growth and development related model components can be combined at an appropriate level of process detail, as dictated by the specific application and the spatial and temporal scales considered (Gaiser et al., 2013). For this study, SIMPLACE included the LINTUL-5 (Wolf, 2012) crop growth and development model and the water balance model DNUNIR from the LINTUL-2 crop growth model (Spitters and Schapendonk, 1990; van Oijen and Leffelaar, 2008). This specific implementation of SIMPLACE considers the effects of climate including limited water supply as described in Spitters and Schapendonk (1990) and Farré et al. (2000), and of limited N supply as described by Shibu et al. (2010). A simple representation of the effects of increased atmospheric CO₂ concentration on biomass production is used by applying the relationship between atmospheric CO₂ and radiation use efficiency, as proposed by Stockle et al.

(1992). The second effect of increased CO₂ considered is the reduction of crop transpiration (i.e. reduction by 10% for all crops when atmospheric CO₂ increases from 370 to 700 $\mu\text{mol CO}_2/\text{mol}$, as based on Ewert et al. (2002) and Kruijt et al. (2008)). The resulting combination of modules enables assessing the impacts of changes in CO₂, temperature, rainfall and crop management (sowing date, varietal characteristics, nitrogen fertilization) on crop yields. Model calibration was performed for units corresponding to the intersection of environmental zones and NUTS2 administrative zones to allow assessment for Europe at NUTS2 level in this study (Wolf et al., 2012; Angulo et al., 2013; Reidsma et al., 2015).

CAPRI (<http://www.capri-model.org/dokuwiki/doku.php>) is a comparative static partial equilibrium model for the agricultural sector, developed for policy impact assessment of the Common Agricultural Policy (CAP) and trade policies from global to regional and farm type scale (Britz et al., 2007; Britz and Witzke, 2012). CAPRI allows the assessment of economic consequences at the regional level across Europe, based on the linkage of its supply and global market module. The *supply module* covers the EU27, Norway, Western Balkans and Turkey, and represents all agricultural production activities and related output generation and input use at a regional or farm type level (Gocht and Britz, 2011) captured by the Economic Accounts for Agriculture, and depicts policy instruments under the CAP. In addition to outputs, including nutrients in manure, the model considers in detail variable costs for different production activities, captures the effects of labour and capital on farmers' decisions and comprises a land market which renders total agricultural land endogenous. Prices are exogenous in the supply module and are provided by the market module. The core of the *market module* consists of a spatial global multi-commodity model for about 50 primary and processed agricultural products, representing the world with approximately 80 country groups or blocks in 40 trading blocks. This module delivers the output prices used in the supply module and allows for market analysis at global, EU and national scale, including a welfare analysis.

FSSIM is a generic bio-economic farm model, developed to simultaneously quantify agricultural, environmental and economic responses of major arable farm types across the EU to new policies and agro-technologies (Janssen et al., 2010; Louhichi et al., 2010). FSSIM simulates farmers' decisions regarding land use and management at farm level in response to changes in policies, technology, yields and product prices. These decisions in turn affect cropping patterns and thereby farm gross income, total costs per farm, farm net income and per farm labour demand, which are the main outputs of FSSIM. FSSIM can assess optimal adaptation strategies to changes in climate and policies for different farm types. FSSIM is generally run for representative farm types in a region, and the results can be aggregated to the regional level.

INTEGRATOR is an environmental agricultural model which quantifies N and greenhouse gas emissions from housing and manure storage systems, agricultural soils, non-agricultural soils and surface waters at EU27 level (de Vries et al., 2011; Kros et al., 2011). For agricultural systems, the calculations of N (NH₃, NO_x, N₂O and N₂) emissions and N leaching to groundwater and runoff to surface water are based on an adapted version of the MITERRA-Europe model (Velthof et al., 2009). This adapted version in INTEGRATOR uses soil emission factors that vary with the N source (e.g. fertilizer and manure types, crop residues and mineralized soil organic N), manure application technique, soil type, land use and precipitation (Lesschen et al., 2011). Modelled losses of N from agricultural systems to groundwater and surface waters include: (i) leaching from stored manure to groundwater, (ii) surface runoff to surface waters, (iii) subsurface runoff to surface waters, and (iv) downward leaching to groundwater, as described in Kros et al. (2012). In this study, INTEGRATOR calculates total N leaching to both ground and surface waters, NH₃ emission and N₂O emission to the air from arable land, which N losses contribute, respectively, to eutrophication, decrease in biodiversity and greenhouse gas accumulation.

2.2. Model linkage and calibration

To answer the policy relevant questions of our test case, the four component models were linked as follows: INTEGRATOR received inputs from FSSIM (for farm and regional analyses) and CAPRI (for regional and EU-scale analyses), while SIMPLACE estimates of yields for future scenarios of climate and CO₂ concentrations were supplied as inputs to each of the other models. Fig. 1 provides an overview of (a) the directional linkage of the four models, (b) the key datasets used, (c) the main input data passed between the models, and (d) the most important outputs (e.g. farm net income, environmental impacts) from the integrated analyses. A key aspect of this model linkage is the harmonization of the N applications and the yields used in the linked models, which requires a common database at regional level for CAPRI, SIMPLACE and INTEGRATOR.

Both FSSIM and CAPRI also calculate various environmental indicators, including some covered by MITERRA-Europe, but typically in less detail. Similarly, CAPRI can provide economic estimates at farm level, however, not at single farm type level, as performed by FSSIM. Similarly, both SIMPLACE and INTEGRATOR account for soil nitrogen amounts, though the approach in INTEGRATOR considers more transformations and processes, whereas SIMPLACE better accounts for nitrogen uptake by crops. Accordingly, the outputs selected from the different models and passed along the model chain reflect their comparative advantages.

Scenarios of changes in CO₂, temperature and rainfall are used by SIMPLACE and further combined with estimates of technological development to assess crop yield impacts. The simulations covered winter wheat, winter rape seed, winter barley, potatoes, sugar beet, grain maize and silage maize. Initial calibration of SIMPLACE for responsiveness to different levels of nitrogen and water availability used agronomic trial datasets for these crops. Further calibration was made for each climate zone (intersection of environmental zone and NUTS2) such that the root mean square error (RMSE) was minimized between (detrended) observed and simulated yields for the period 1983–2006 for each of the simulated crops to derive a time-invariant correction factor between observed and simulated yields (Angulo et al., 2013). The calibration was performed using nitrogen inputs (fertilizer, crop residues and animal manures) for each NUTS2 regions (Nomenclature of Territorial Units for Statistics, http://epp.eurostat.ec.europa.eu/portal/page/portal/nuts_nomenclature/introduction, i.e. administrative regions in EU27) from the CAPRI database. Future crop yields were simulated for

a 24-year period centred on 2050 for the different climate change scenarios described below. The future yield changes from SIMPLACE due to climate and CO₂ were then adjusted by including expected yield increases due to technology as estimated from historical yield trends with modified growth rates based specific to each scenario (Ewert et al., 2005). The method to calculate the historical trends for each NUTS2 considered the trends at each of the NUTS2, country and EU-27 grouping levels, as weighted by the strength of their respective regressions. The resulting crop yield changes to 2050 were used by CAPRI in combination with scenario specific economic drivers at the global scale (e.g. population and income growth, taking from recent AGMIP runs, Nelson et al., 2014) and matching assumptions of agricultural trade policies, to simulate impacts on global agricultural markets, including price changes. At the global scale, yield changes were based on a study on the climate change impacts on global crop production by Parry et al. (2004).

As CAPRI, FSSIM and INTEGRATOR cover many more crops than the seven simulated with SIMPLACE, the yield responses for these seven crops were extrapolated to others by applying a crop mapping table (e.g. mean yield change for wheat and barley have been used for all other grain crops). These yield changes along with prices simulated in CAPRI were used by FSSIM to calculate future changes in farm net income, farm labour demand and cropping patterns for different future scenarios. SIMPLACE results of future yield changes together with simulated changes in cropping patterns from FSSIM were then used by INTEGRATOR to calculate N applications, uptake and losses to air and water for the main crop types in each region and scenario (see Fig. 1). In summary, the following outputs from the linked models were considered in the study: future changes in cropping patterns, farm gross and net income, labour demand, N applications and N emissions at the farm and regional level.

2.3. Evaluated scenarios for agriculture in 2050 with different combinations of changes in climate, technologies and prices

The linked models were used first in the base period (2003–2005), in particular for calibrating FSSIM to the observed cropping patterns, and next, for three socio-economic and emission scenarios, including a baseline scenario for 2050 (B1), a strong economic growth scenario (A1-b1) and a weak-economic growth scenario (B2) for 2050 (Table 1), as described by IPCC (2007). The A1-b1 scenario assumes

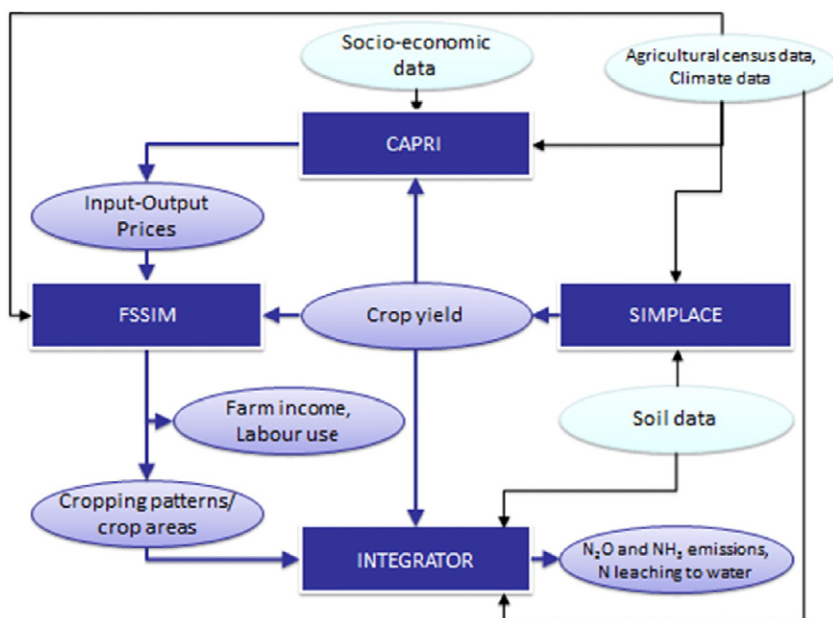


Fig. 1. Main flows of inputs and outputs to and from the four models that are linked for the Agri-test case.

Table 1

Description of the scenarios with their assumptions and related data as partly derived from the Agri-Adapt project (Source: Wolf et al., 2012; Reidsma et al., 2015). Note that for these scenarios all the different types of changes towards 2050 (i.e. changes in climate, technology, management, prices and policies) are indicated. Hence, this description applies to the CLIMTP variant (see the text for more information about the variants).

	Base year [2004]	B1 (baseline) [2050]	B2 [2050]	A1-b1 [2050]
Exogenous assumptions	Observed data (average 2003–2005) taken from EuroStat, FAO, OECD etc.	Inflation rate of 1.9% per year Constant exchange rates GDP and population growth compared to 2003–2005 are derived from IMAGE 2.1 model (de Vries et al., 2000) results as prepared for the Fourth IPCC assessment report ^a . B2 implies a decrease and A1-b1 an increase of agricultural demand compared to the baseline	Simulation results (European Commission and IFPRI)	
Commodity prices	Observed prices (average 2003–2005)	Extrapolated from market outlooks (European Commission and IFPRI)		
Input prices	Observed prices (average 2003–2005)	Extrapolated from market outlooks for baseline. For B2 and A1-b1, changes in oil prices are estimated from changes in GDP and population growth, and mapped into changes in input prices according to their energy cost share		
Yield	Observed yields (average 2003–2005)	SIMPLACE simulation (BCCR_BCM2_0/SRES B1)	SIMPLACE simulation (Pattern-scaled SRES B2 15-model ensemble mean)	SIMPLACE simulation (SRES A1B 15-model ensemble mean)
Set-aside and quota policies	With obligatory set-aside and quota (milk and sugar)	Abolishing obligatory set-aside, expiry of sugar and milk quota		
Premium scheme	2003 CAP reform (decoupled + partially coupled payment) ^b	2009 Health Check (decoupled payment, increased modulation) ^c		
WTO trade policy	Tariffs and TRQ as in 2004	Tariffs and TRQ as in 2004		Reduction of tariffs and expansion of TRQ (sensitive products) ^d

^a IPCC (2007).

^b Since 1992, the common agricultural policy (CAP) has been reformed, aimed at moving away from a policy of price and production support to a more comprehensive policy of farmer income support. We use the Single Farm Payment as operated until 2013 as a proxy for the probable support to agricultural land under the CAP in the year 2050, neglecting any detailed implementation issues as e.g. underlying the 2013 CAP reform (greening measures, shifts between Pillars I and II, etc.). For more information about the CAP reform 2003, decoupling, modulation, etc., see http://ec.europa.eu/agriculture/mtr/sum_en.pdf.

^c On 20 November 2008 the EU agriculture ministers reached a political agreement on the Health Check of the CAP. The agreement abolishes arable set-aside. Ministers also agreed to increase modulation, whereby direct payments to farmers are reduced. For more information see http://ec.europa.eu/agriculture/healthcheck/index_en.htm.

^d During WTO negotiations (2009) Ambassador Crawford Falconer, chairperson of the agriculture negotiations, circulated his latest revised draft “modalities” text. This text contained formulas for cutting tariffs and trade-distorting subsidies and related provisions. See for more information: http://www.wto.org/english/tratop_e/agric_e/chair_texts08_e.htm.

rapid economic growth, more free global trade and a strong increase in wealth and thus food expenditures until 2050, whereas the B2 scenario assumes more limited economic growth, more restrictive trade policies and environmental taxes, and thus a more limited increase in wealth and food expenditures. The B1 scenario represents a possible future development where developments are assumed to be roughly in between those for the A1-b1 and B2 scenarios.

We have used here the so-called SRES scenarios and not the more recent Representative Concentration Pathways (RCPs) scenarios. van Vuuren and Carter (2014) have performed a comparison between the SRES and the RCPs with respect to socio-economic pathways and projected atmospheric composition, radiative forcing and climate characteristics. Although the relationships between these different sets of scenarios are complex, they conclude that A1-b1 is roughly similar to the RCP 6.0 scenario, the B1 scenario roughly similar to the RCP 4.5 scenario, and that the B2 scenario is positioned in-between the RCP 4.5 and the RCP 6.0 scenarios.

In the integrated analyses, the aim was to assess climate change impacts in the context of technological, socio-economic (markets) and policy changes to 2050. This is relevant as climate change is only one of the drivers of agricultural systems in 2050, and climate change depends on the development of the global economy with resultant impacts on future CO₂ concentrations. The main factors that influence future agricultural systems and their development and that differ between the three scenarios for 2050 (as specified in Table 1), are:

- *Changes in climate conditions* (as based on the three IPCC scenarios) and CO₂ concentrations; climate data were derived from the Intergovernmental Panel on Climate Change (IPCC) Data Distribution Centre (DDC IPCC, 2010) and clearly differ between the scenarios, as described in Wolf et al. (2012, p. 16). Atmospheric CO₂ which increases over time and affects future crop yields, was set at 369 μmol CO₂/mol for base year and 488, 532 and 478 μmol CO₂/mol for B1, A1-b1 and B2 scenarios, respectively (IPCC, 2001). These data were used in SIMPLACE to simulate crop yields.

- *Technological improvement in crop varieties and management* which may lead to higher yields in the future; these improvements are assumed to be related to both future changes in global demand of agricultural commodities and in investments to improve crop varieties and management, dependent on the different IPCC scenarios; in this study, the impact of technological change on future yields has been based on extrapolating historical yield trends to 2050 differently for each scenario (Ewert et al., 2005); these yield changes are added to the yield changes due to climate change, as calculated by SIMPLACE over Europe.
- *Changes in the prices of the inputs for agricultural production and the agricultural products*; CAPRI simulates price changes for agricultural inputs and products, evolving from the shocks defined in the different IPCC scenarios with regard to changes in GDP, population, trade and CAP policies and yield changes resulting from climate change. The simulated prices equilibrate again demand and supply for agriculture products globally.

The impacts of these three factors, which occur simultaneously in the real world, were examined individually to gauge their relative importance by decomposing the scenarios considered in CAPRI, FSSIM and INTEGRATOR into the following variants:

1. CLIM – climate change only analysis. Projecting expected climate change impacts in 2050 on present farming systems in NUTS2 regions, with their present layout, agro-management and productivity, markets and policies. We used data from the base year (Table 1) and only changed the yields due to climate and CO₂ change for three scenarios for 2050 and their related N applications. Note that the policy changes (see Table 1) are essentially the same for the three scenarios. We assume that the impacts of the CAP towards the future on EU farming systems will become more and more limited due to liberalization and globalization of agricultural markets and that EC policies will develop practically in the same way under the three scenarios. However, different macro-economic developments (e.g. changes in

Table 2

Overview of the applied scenarios and types of analysis (as indicated by the term: *variants*) with, respectively, inclusion of the effects of increased atmospheric CO₂ and climate change on crop yields only (CLIM), additional inclusion of the effects of improved technology and management on crop yields (CLIMT), both CLIM and the effects of changes in prices (CLIMP), and both CLIM and the effects of all factors together (CLIMTP) for 2050. Policy changes are similar in the three scenarios, but different macro-economic developments as based on GLOBIOM modelling (see <http://www.globiom.org/>), are used per scenario.

Scenarios	Variants	Climate change (change in CO ₂ and climatic parameters)	Technological change (e.g. improved cultivars)	Price changes for both inputs and products
B1, A1-b1, B2	CLIM	Yes	No	No
B1, A1-b1, B2	CLIMT	Yes	Yes	No
B1, A1-b1, B2	CLIMP	Yes	No	Yes
B1, A1-b1, B2	CLIMTP	Yes	Yes	Yes

GDP and food consumption at the global scale, as derived from Nelson et al., 2014) were used for the different scenarios.

2. CLIMT – climate change plus improved technology: as CLIM, but projecting the historical trends (as modified for each scenario) in yield improvement (due to improvements in crop breeding and management) onto the crop yields simulated by SIMPLACE.
3. CLIMP – climate change plus the effects of changes in the prices for inputs and agricultural products, as simulated with CAPRI for the relevant scenario.
4. CLIMTP – the effects of all factors together, i.e. 2050 integrated analysis.

The assessments were carried out for the base period (2003–2005) and for 12 different combinations of the three scenarios (baseline B1, A1-b1 and B2) and four variants (CLIM, CLIMT, CLIMP and CLIMTP) to 2050, as shown in Table 2. Whereas yield and market changes are applied at European and global scale, we concentrate in our analysis on five considerably different (with respect to farming and environmental conditions) regions/countries over Europe, i.e. Denmark as a whole; Flevoland, The Netherlands; Midi Pyrenées, France; Zachodniopomorski, Poland and Andalucia, Spain.

2.4. Input data

The main input data used in this study, the main flows of data between the models, and the main results from the linked modelling approach are presented in Fig. 1.

2.4.1. SIMPLACE

The main input data are weather data, soil and crop data, and N supply data. Historical weather and future scenario climate data used were obtained from the SEAMLESS database (van Ittersum et al., 2008) for 533 climate zones in EU25 (Janssen et al., 2009; Andersen et al., 2010) for the period 1983–2006. A climate zone is a spatial unit that combines NUTS2 regions and Environmental Zones (EnZ) (Metzger et al., 2005). Data included daily rainfall, maximum and minimum air temperature, global solar radiation, and wind speed.

Soil characteristics at the level of AgriEnvironmental Zones (AEnZ) (Hazeu et al., 2010) were available from the Pan European SEAMLESS database (van Ittersum et al., 2008; Andersen et al., 2010). In this study, only the dominant soil type per AEnZ (i.e. covering the largest area in each AEnZ) was considered and aggregated to the level of NUTS2 regions, for which yield statistics were also available. N supply data are supplied by CAPRI.

Yearly sowing and harvest dates for the crops simulated with SIMPLACE were obtained from the JRC/MARS Crop Knowledge Base for 233 NUTS2 regions across Europe (JRC, 1998), averaged to the level of 13 Environmental Zones (EnZ) and subsequently disaggregated again to the climate zones. Yearly yields were available for NUTS2 regions from 1983 to 2006 from the CAPRI database.

Total crop available nitrogen was determined from historical data in the CAPRI database. The data used consisted on nitrogen applied annually from various sources (mineral fertilizer, manure and crop residues) for each NUTS2 region and crop. Mineral N-fertilizer was reported per crop per NUTS2 and used directly. Plant available nitrogen from manure was determined as the sum of mineral-N and the available fraction of organic-N in the manure. For the scenarios runs, nitrogen levels were increased for each scenario and variant as related to both the changes in water-limited yields due to climate and CO₂ change and a moderate increase in N use efficiency. EUROSTAT statistics (2010) on irrigated areas were used to identify the proportion of each crop in each NUTS2 that is

Table 3

Main input data for FSSIM modelling of arable farms and the applied data links to SIMPLACE, CAPRI and INTEGRATOR.

Management per crop type Data are given for the main crop activities in the region for the base year, as collected by local experts		Changes per crop type and data link Relative changes based on input data from CAPRI (C) and SIMPLACE (S); data link to INTEGRATOR (I) for N emission calculations	
Input data	Unit	Input data	N emissions, in kg N/ha
Yield	ton/ha	Relative change in crop yield (S)	NH ₃ emission (I)
By product	ton/ha	Relative change in by-product (S)	N ₂ O emission (I)
Price of yield	euro/t	Relative change in price of yield (C)	NO _x emission (I) ^a
Price of by-products	euro/t	Relative change in price of by-product (C)	N ₂ emission (I) ^a
Amounts of active biocides applied	kg/ha	Relative change in costs (C); Change in costs as related to yield changes	N leaching to ground and surface water (I)
Amounts of fertilizer N, P and K applied ^b	kg/ha	Constraints and other input data per farm type Base year data mainly based on FADN; Subsidies are the same for base year, CLIM and CLIMT and become zero for CLIMP and CLIMTP with price changes; resources (i.e. available land, water and labour) per farm type are similar for base year (data from FADN) and the three scenarios and four variants	
Costs of biocides applied	euro/ha	Input data	Unit
Costs of fertilizer nutrients applied	euro/ha	Total labour available	hours
Other variable costs	euro/ha	Total land area	ha
Irrigation water applied	m ³ /ha	Total irrigable land area	ha
Labour demand	hour/ha	Water available	m ³
		Subsidies	euro
		Actual cropping pattern	ha
		Available crop rotations	–

^a This N emission is not shown in the reported outcomes.

^b These N applications are not used by FSSIM in this study, but N applications are calculated in the INTEGRATOR model on the basis of the yields and are harmonized with those used in SIMPLACE and CAPRI.

Table 4

Main input data for INTEGRATOR model and the applied data links to SIMPLACE and FSSIM.

N excretion, emission and uptake factors Data depend amongst others on N source, soil type, land use and precipitation		Changes per crop type, in cropping pattern and data link Relative changes based on input data from SIMPLACE (S) or FSSIM (F)	
Input data	Unit	Input data	Unit
N excretion rate	kg N/animal	Relative change in crop yield (S)	–
N uptake factor ^a	–	Change in cropping patterns (F)	ha
N content in crops	–	Other input data	
		Based mainly on FAO data	
		Input data	Unit
NH ₃ emission factor	–	Land cover	ha
N ₂ O emission factor	–	Land use (crops in arable land)	ha
NO _x emission factor	–	Crop yields	kg/ha
N leaching factors	–	N fertilizer application	kg/ha
N surface runoff factor	–	Animal livestock numbers	–
N subsurface runoff factor	–	Total N deposition levels	kg/ha
		Soil properties	

^a Efficiency factor of the effective N applied.

irrigated. This proportion was used to calculate weighted averages of the yield between the nitrogen and water limited yields (no irrigation) and nitrogen limited yields (full irrigation) for each crop.

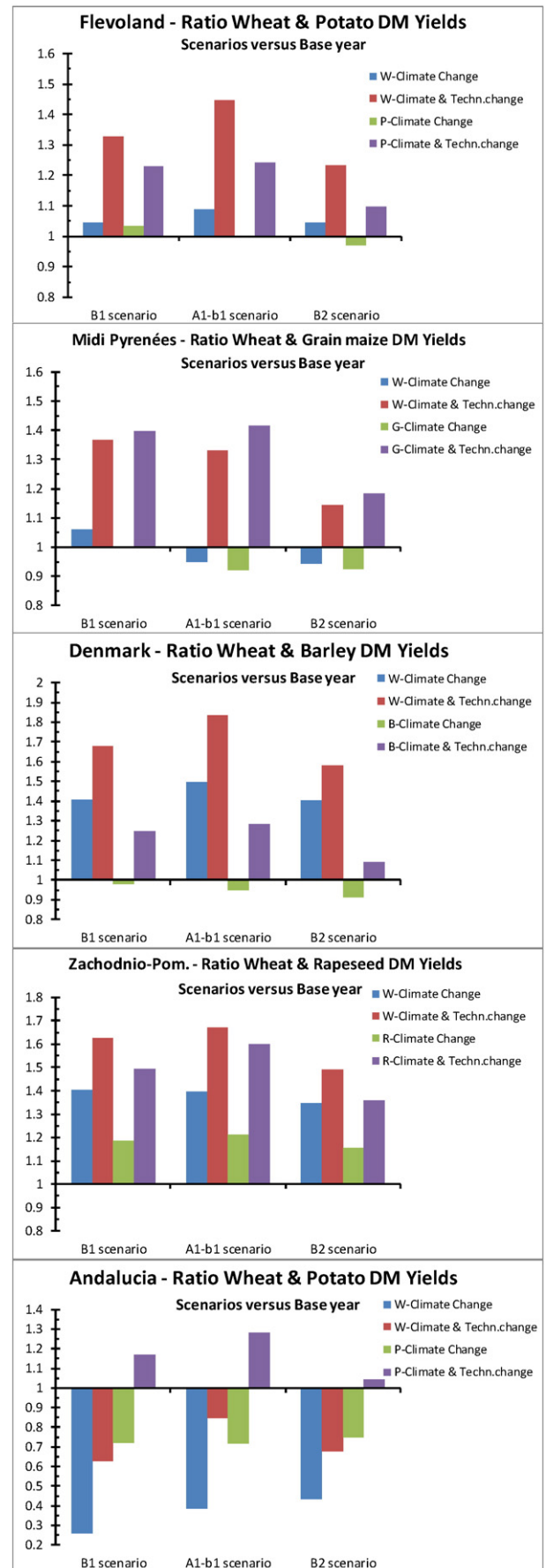
2.4.2. CAPRI

Apart from socio-economic input data, the main agricultural input data for CAPRI include land cover/land use (grass and crops in arable land), crop yields, and animal livestock numbers, which are mainly based on EUROSTAT statistics at regional level. They are combined with engineering data to e.g. derive N fertilizer application, while appropriate transformations ensure consistency to national and global scale. A further disaggregation took place in CAPRI-Spat to about 100,000 1 × 1 km grid cells, with distinction in irrigated and non-irrigated land in case of crop yield and levels. That disaggregation which also covers animal numbers and N applications is described in detail in Leip et al. (2008).

2.4.3. FSSIM

Table 3 reports the main input data for FSSIM along with the data linkage to SIMPLACE (i.e. yield changes), CAPRI (i.e. price changes for inputs and products) and INTEGRATOR (crop–soil area fractions). The main types of data used by FSSIM are the current agricultural management data, the available resource endowments for each farm type (note that currently available land, water and labour are also used for the future scenarios and variants), and the changes for the different scenarios towards 2050 (Table 3). The management information consists of the crop activities (i.e. main crop rotations) in a region with their yields, required inputs and their costs for average management. These data refer to the base year (2003–2005) and were collected for each crop–soil combination in a region based on surveys for these years (Zander et al., 2009). FADN data (Farm Accountancy Data Network, see <http://ec.europa.eu/agriculture/rca/>) were used to define the available resource endowments per farm type. Changes towards the future according to the three scenarios are described in Table 3 and in Section 2.3.

Fig. 2. Relative changes in crop yields of two main crops per region in Flevoland (winter soft wheat (W) & seed potato (P)), Midi Pyrenées (winter durum wheat (W) & grain maize (G)), Denmark (winter soft wheat (W) & barley (B)), Zachodniopomorski (winter soft wheat (W) & rape seed (R)), and Andalucia (wheat (W) & potato (P)) for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering first, the effects of climate change and increased atmospheric CO₂ on crop yields only (CLIM) and next, the effects of both CLIM and technology and management improvements on crop yields (CLIMT).



2.4.4. INTEGRATOR

Table 4 reports the main input data for INTEGRATOR and the data linkage to SIMPLACE (i.e. yield changes) and FSSIM (crop–soil area fractions). The main types of data used by INTEGRATOR are (i) N process factors, (ii) data on land cover, land use, and animal numbers which all change over time, and on soil properties which remain constant and (iii) changes in crop yield and cropping pattern which are based on SIMPLACE and FSSIM outputs, respectively. N process factors include N excretion, N uptake, N emission, N leaching and N runoff fractions (Table 4); these factors depend on e.g. animal categories, land use, crop type, soil type and climatic factors. For example, N_2O emission factor is a function of N source, application technique, soil type, land use and amount of precipitation, as based on Lesschen et al. (2011). NH_3 emission factor is a function of housing and manure storage systems and of manure and fertilizer types, while the sum of the N leaching and subsurface runoff factors is related to soil type, land use, soil organic content, precipitation surplus, temperature and rooting depth (Velthof et al., 2009). We assume that the N use efficiency in INTEGRATOR increases by 20% towards 2050 (i.e. increase in N uptake factor in Table 4) and also that the housing systems and manure application techniques improve over time, leading to reduced NH_3 emissions.

Regional cropland area data in INTEGRATOR are based on CORINE 2000 (ETC, 2000), whereas the land share data per crop type are based on CAPRI-SPAT (Leip et al., 2008). We have assumed here that there are no changes in total cropland area towards 2050, but only in the cropping patterns, to be able to interpret the results. Current data on N fertilizer use and animal livestock numbers in INTEGRATOR are based on the FAO database (FAO, 2007), whereas results from IMAGE modelling exercises for the different scenarios are used to scale these current data towards the future. These changes have affected the ratio between N fertilizer and manure inputs. CAPRI-SPAT data were used for the downscaling of the output data. Used soil type, texture class, C content and C/N ratio data were derived from the up-scaled SPADE/WISE database (Heuvelink et al., 2009).

3. Results at regional scale

In the following sections, results are presented in terms of relative changes compared to the base year (2003–2005) for (i) agronomic indicators and product prices, i.e. crop yields, prices of agricultural products and cropping patterns, (ii) socio-economic indicators, i.e. farm gross income, farm net income and farm labour demand, and (iii) environmental indicators, i.e. N losses to air and water, for each scenario and variant. Changes are qualitatively described using slight, moderate, considerable, strong and very strong, to denote relative changes of 0–20%, 20–40%, 40–60%, 60–80% and >80%, respectively.

Cropping patterns are reported in detail as they explain to a large extent, together with the exogenous changes in crop yield, the economic and environmental results. Farm income is influenced by both (1) changes in revenues per hectare for each crop as a result of the exogenous yield and price changes, as well as (2) the subsequent endogenous adjustment of cropping patterns (e.g. shift from grain crops to a high value crop as seed potato). The same applies to changes in N input at farm level, which reflects the crop nutrient needs at given yields for the different crops, which in turn largely affects the N emissions to air and water.

3.1. Changes in agronomic indicators and product prices

3.1.1. Crop yields

Crop yields mostly increase due to climate change towards 2050, mainly due to higher atmospheric CO_2 . These increases tend to become much larger if changes in farm technology (e.g. improved crop varieties) and management are also considered (Ewert et al., 2005). The yield changes are specific per crop type and region but, in general, they are slightly to moderately positive in Northern and Central Europe when

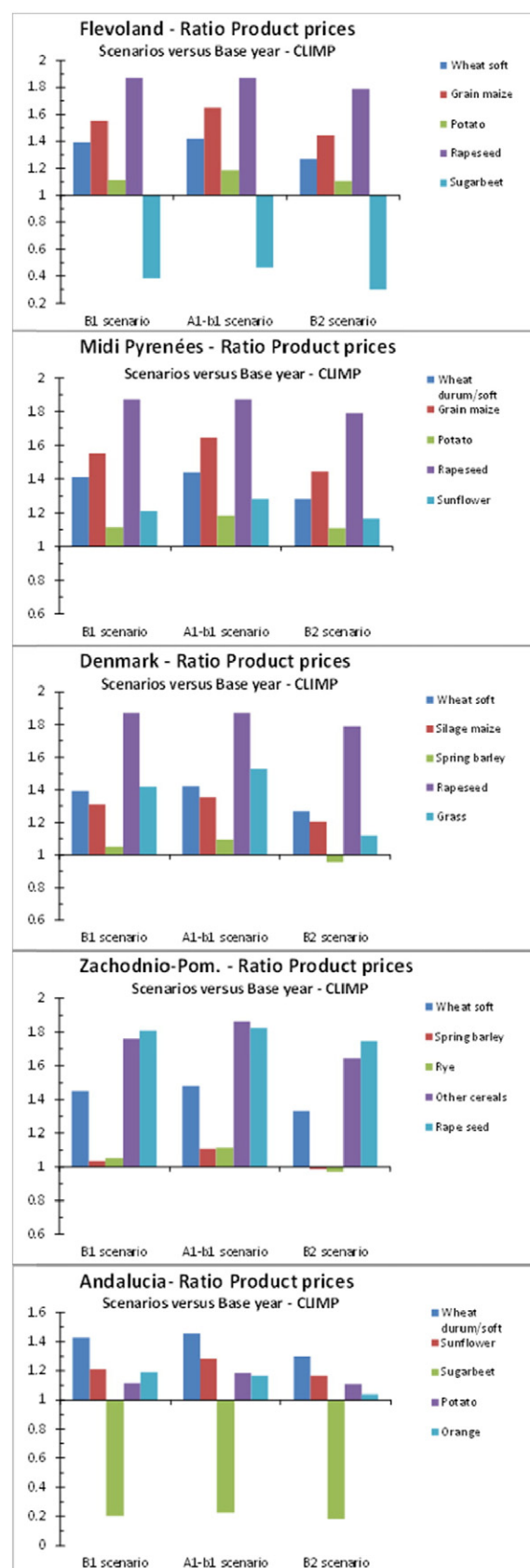


Fig. 3. Relative changes in inflation-corrected (assuming inflation rate of 1.9% per year) prices of the main crop products on mean farms in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalusia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005) as computed with the CAPRI model – no technology change & variant CLIMP.

only changes in climate are considered and moderately to strongly positive for the combination of climate and technology change. In Southern Europe (e.g. Andalusia) the yield changes are strongly to moderately negative when only changes in climate are considered and moderately negative to slightly positive for the combination of climate and technology change. This underlines that in the long term, under European conditions, technology development can be considered at least as important as climate change for crop yield developments.

As an example, crop yield changes are presented in Fig. 2 for the two most representative crop types per region, i.e., seed potato and winter wheat in Flevoland, winter durum wheat and grain maize in Midi-Pyrénées, winter soft wheat and barley in Denmark, winter wheat and rape seed in Zachodniopomorski, and wheat and potato in Andalusia. Yield changes due to climate change are apparently less positive or more negative for crops in Southern Europe than for crops in the other parts of Europe, as particularly shown for Andalusia and to a lesser extent for Midi Pyrénées (Fig. 2). This is due to the increasing risk for severe drought and high temperature periods in Southern Europe. Results also show that the yield increases for a situation with both climate and technology changes are stronger for the baseline (B1) and the strong-economic growth A1-b1 scenarios than for the low economic growth B2 scenario, assuming weaker food demand and lower investment in improved crop varieties for the B2 scenario.

3.1.2. Prices of agricultural products

Product prices change from the base period to 2050 to differing extents, depending on the crop type and the scenario. These price changes, which affect cropping patterns choices for the different scenarios (Section 3.1.3), are given for the main crop types cultivated in the five regions. The variant CLIMP shows the price changes from CAPRI resulting when no farm technology and management improvements are assumed (Fig. 3). The product prices which are inflation corrected based on an inflation rate of 1.9% per year, appear mainly to increase by a factor between 1.1 and 1.8 from 2004 to 2050. These rather large price increases reflect the fact that population and income increase in the scenario without any technical progress in agriculture. The increases are strongest for grain maize and particularly rape seed and smallest for potato, rye and fodder crops like silage maize and grass. For sugar beet the disappearance of the sugar quota is expected to lead to a strong price reduction over time. Product prices show relatively small differences between the three scenarios. This reflects the fact that an increase in income leads to limited changes in primary product demand. Next, the price changes are derived from CAPRI simulations for a situation in which farm technology (e.g. improved crop varieties) and management improvements are assumed (see Appendix, Fig. A-1; variant CLIMTP). The product prices appear mainly to change by a factor between 0.9

and 1.5 from 2004 to 2050. These smaller price increases compared to those for variant CLIMP (Fig. 3) are due to the stronger increases in crop production with farm technology and management improvements (Fig. 2).

3.1.3. Cropping patterns

Cropping patterns change for each combination of scenario and variant. To give insight into these changes, Fig. 4 presents the cropping patterns for typical farms in Flevoland, Midi Pyrénées, Denmark, Zachodniopomorski and Andalusia for the base year, whereas the cropping patterns in 2050 for the four variants of the B1 scenario are given in Table 5. Such changes were modelled with the FSSIM model which simulated farmer's decisions regarding land use, and were the result of the interacting effects of climate change, technology and management improvements and price changes (see Section 2.3). Finally, Fig. 5 presents the cropping patterns for the three scenarios and four variants in 2050 for Flevoland as an example.

In Flevoland, the main crops in the base year are seed potato (37%), onion (16%), sugar beet (15%) and winter wheat (13%), with smaller area fractions for spring wheat and spring barley. In Midi Pyrénées, the main crops are winter durum wheat (22%), grain maize (21%) and sunflower (20%), with smaller area fractions for winter soft wheat, fallow land and peas. In Denmark the main crops in the base year are winter soft wheat (32%) and spring barley (29%) with smaller area fractions for winter barley, fallow land, and rape seed (see Fig. 4 and Table 5). In Zachodniopomorski the main crops in the base year are winter soft wheat (25%), spring barley (13%), rape seed (12%), rye (12%) and other (mixed) cereals (11%), with smaller area fractions for triticale, oats and spring wheat. In Andalusia, the main crops in the base year are cotton (25%), durum wheat (21%) and sugar beet (14%), a considerable area fraction is fallow land (24%), with a smaller area fraction for ware potato.

In Flevoland, the B1 scenario for 2050 results in a strong increase in the area fraction with seed potato (i.e. 52%) compared to that in the base year (Fig. 4), at the cost of the area fractions for onion and spring wheat in case of climate and CO₂ change only, CLIM (Fig. 5; top graph). The A1-b1 and B2 scenarios result in less specialization in seed potato and more specialization in onions and winter soft wheat. The changes in the cropping pattern per scenario due to technology and management improvements (CLIMT), compared to those from CLIM, result in a strong specialization in seed potato (between 52% and 65%) for the three scenarios (Fig. 5; second graph), which is due to the moderate yield increases by technology change in Flevoland (Fig. 2). The price changes in 2050 (i.e. CLIMP versus CLIM) result for the A1-b1 and B1 scenarios in very strong specialization in onion (from 7 to 17% for CLIM to around 40%) at the cost of the seed potato area (from 34 to 52% at CLIM to

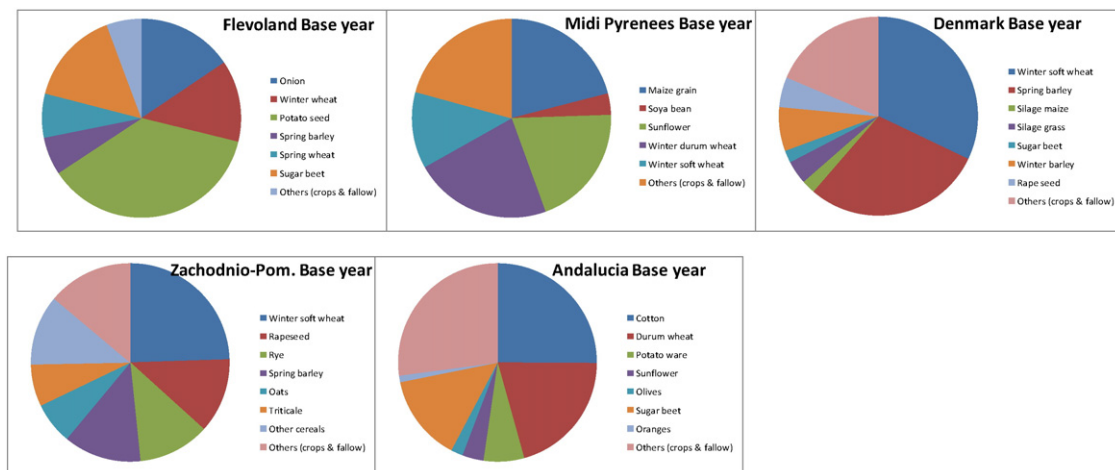


Fig. 4. Cropping patterns for typical farms in Flevoland, Midi Pyrénées, Denmark, Zachodniopomorski and Andalusia for the base year (2003–2005).

Table 5

Relative crop area fractions (—) for base year (2003–2005) and baseline (B1) scenario for 2050 in the regions Denmark as a whole, Flevoland, Midi-Pyrénées, Zachodniopomorski and Andalucía with CLIM, CLIMT, CLIMP and CLIMTP variants, respectively. Note that the cropping patterns are based on aggregating the results from the different farm types in that region weighted according to the number of represented farms.

Region (no. of farm types)	Base year	B1 scenario CLIM	B1 scenario CLIMT	B1 scenario CLIMP	B1 scenario CLIMTP
<i>Flevoland (2)</i>					
Maize fodder	0.02	0.01	0.00	0.01	0.00
Maize grain	0.00	0.00	0.00	0.00	0.00
Onion	0.16	0.07	0.05	0.38	0.14
Peas	0.01	0.00	0.00	0.00	0.00
Potato seed	0.37	0.52	0.65	0.36	0.46
Potato ware	0.01	0.01	0.00	0.01	0.01
Rape seed	0.00	0.00	0.00	0.00	0.00
Spring barley	0.06	0.06	0.07	0.05	0.09
Spring wheat	0.07	0.03	0.00	0.04	0.00
Sugar beet	0.15	0.15	0.11	0.08	0.09
Tulip	0.00	0.00	0.00	0.01	0.00
Winter wheat	0.13	0.12	0.11	0.05	0.18
Fallow	0.02	0.02	0.02	0.02	0.02
Total area (ha)	60.7	60.7	60.7	60.7	60.7
<i>Midi-Pyrénées (3)</i>					
Maize grain	0.21	0.19	0.60	0.66	0.68
Maize fodder	0.01	0.00	0.00	0.01	0.01
Oats	0.00	0.00	0.00	0.00	0.00
Peas	0.04	0.04	0.02	0.02	0.02
Rape seed	0.02	0.02	0.01	0.04	0.04
Soya bean	0.03	0.03	0.01	0.00	0.00
Sunflower	0.20	0.19	0.06	0.08	0.05
Winter barley	0.03	0.03	0.01	0.01	0.01
Winter durum wheat	0.22	0.26	0.10	0.00	0.00
Winter soft wheat	0.12	0.13	0.07	0.13	0.15
Fallow	0.11	0.11	0.11	0.06	0.06
Total area (ha)	102.0	102.0	102.0	102.0	102.0
<i>Denmark (8)</i>					
Alfalfa fodder	0.00	0.00	0.00	0.00	0.00
Grass seed	0.03	0.03	0.02	0.03	0.03
Grass fodder	0.04	0.11	0.06	0.25	0.19
Maize fodder	0.02	0.08	0.08	0.10	0.08
Oats fodder	0.03	0.03	0.02	0.02	0.02
Peas fodder	0.01	0.01	0.01	0.00	0.00
Potato ware	0.02	0.02	0.03	0.02	0.02
Rape seed	0.05	0.02	0.02	0.02	0.03
Rye	0.03	0.01	0.00	0.00	0.00
Spring barley	0.29	0.08	0.12	0.02	0.03
Sugar beet	0.02	0.02	0.05	0.01	0.01
Triticale fodder	0.00	0.00	0.00	0.00	0.02
Winter barley	0.07	0.03	0.03	0.01	0.01
Winter soft wheat	0.32	0.51	0.50	0.50	0.54
Fallow	0.06	0.06	0.06	0.01	0.01
Total area (ha)	64.6	64.6	64.6	64.6	64.6
<i>Zachodniopomorski (7)</i>					
Maize fodder	0.02	0.02	0.02	0.02	0.03
Oats	0.07	0.06	0.04	0.01	0.01
Peas	0.01	0.01	0.01	0.01	0.01
Potato ware	0.03	0.03	0.03	0.03	0.04
Rape seed	0.12	0.14	0.14	0.26	0.22
Rye	0.12	0.10	0.09	0.01	0.02
Spring barley	0.13	0.04	0.04	0.00	0.00
Spring wheat	0.06	0.00	0.00	0.00	0.00
Sugar beet	0.01	0.01	0.01	0.00	0.00
Triticale	0.07	0.07	0.07	0.09	0.06
Winter soft wheat	0.25	0.42	0.43	0.38	0.37
Other cereals	0.11	0.10	0.11	0.18	0.23
Fallow	0.00	0.00	0.00	0.00	0.00
Total area (ha)	40.1	40.1	40.1	40.1	40.1
<i>Andalucía (8)</i>					
Cotton	0.25	0.27	0.25	0.28	0.28
Olives	0.02	0.03	0.03	0.05	0.05
Oranges	0.01	0.00	0.00	0.04	0.06
Potato ware	0.07	0.05	0.07	0.06	0.07
Rice	0.00	0.00	0.00	0.00	0.00
Sugar beet	0.14	0.13	0.14	0.06	0.04

Table 5 (continued)

Region (no. of farm types)	Base year	B1 scenario CLIM	B1 scenario CLIMT	B1 scenario CLIMP	B1 scenario CLIMTP
Sunflower	0.03	0.04	0.04	0.04	0.04
Vineyard	0.01	0.01	0.00	0.01	0.01
Durum wheat	0.21	0.19	0.20	0.17	0.18
Soft wheat	0.02	0.02	0.02	0.02	0.02
Fallow	0.24	0.26	0.24	0.25	0.24
Total area (ha)	36.7	36.7	36.7	36.7	36.7

30–36%), and result for the B2 scenario in very strong specialization in seed potato area (from 25% for CLIM to 57%) at the cost of area for onion and sugar beet (Fig. 5; third graph). The changes in both prices and technology and management (CLIMTP) for 2050 result for the three scenarios also in a strong specialization in seed potato area (between 46% and 55%; Fig. 5; bottom graph).

In Midi Pyrénées, the area fraction of durum wheat slightly increases for the B1 scenario compared to those in the base year (i.e. from 22% to 26%), which results in limited decreases in grain maize and sunflower areas in case of climate and CO₂ change (CLIM) only (Table 5). Technology and management improvements (CLIMT), compared to those from CLIM, result in a strong specialization in grain maize (i.e. 60%) at the cost of the areas with all other crops. For the A1-b1 and B2 scenarios the same specialization in grain maize is observed for the CLIMT variant (not shown). The price changes for 2050 both with and without technology change (i.e. CLIMP and CLIMTP versus CLIM) also result, across the three scenarios, in a substantial increase in area for grain maize (between 63% and 68%) at the expense of all other crop areas (see Table 5 for B1 scenario).

In Denmark, the B1 scenario results in strong increases in area fractions for winter soft wheat, fodder grass and fodder maize compared to those for the base year, at the expense of the spring barley area fraction in case of climate and CO₂ change (CLIM) only (Table 5). For the A1-b1 and B2 scenarios the changes in cropping pattern are similar to those for the B1 scenario (not shown). The area distributions for CLIMT are essentially the same as those for CLIM for the three scenarios (see Table 5 for B1 scenario). The price changes for 2050 (i.e. CLIMP versus CLIM) result, across scenarios, in very strong and moderately strong increases in areas with fodder grass and fodder maize, and in very strong decreases in areas with spring barley and fallow (see Table 5 for B1 scenario). The price and technology changes to 2050 (i.e. CLIMTP versus CLIM) result, across scenarios, in the same area changes as from price changes to 2050 alone (see Table 5 for B1 scenario).

In Zachodniopomorski, the B1 scenario results in a strong increase in area fraction for winter soft wheat compared to those for the base year, at the expense of the spring barley and spring wheat area fractions in the case of climate and CO₂ change (CLIM) only (Table 5). For the A1-b1 and B2 scenarios, the changes in cropping pattern are similar to those for the B1 scenario (not shown). The specialization in winter soft wheat is similar (i.e. 43%) for CLIMT and CLIM for the three scenarios (see Table 5 for B1 scenario). The price changes for 2050 (i.e. CLIMP versus CLIM) result in specialization in both winter soft wheat (i.e. 38%), rape seed (i.e. 26%) and other (mixed) cereals (i.e. 18%) at the cost of the areas with most other crops (see Table 5 for B1 scenario), for each of the scenarios considered. The price and technology changes for 2050 (i.e. CLIMTP versus CLIM) result, again across scenarios, in the same area changes as for price changes alone (see Table 5 for B1 scenario).

In Andalucía, climate and technology changes (CLIM and CLIMT) result in essentially no changes in area fractions compared to those for the base year (see Table 5 for B1 scenario), across all future scenarios. The price changes to 2050, with and without technology changes (i.e. CLIMP and CLIMTP versus CLIM) result in considerably and slightly increasing areas for oranges and olives, respectively, at the cost of the strongly decreasing sugar beet area (see Table 5 for B1 scenario). This is consistent across the scenarios.

3.2. Changes in socio-economic indicators

The socio economic indicators used in our analysis are farm gross income (i.e. yields times product prices for the cultivated crops), farm net income (i.e. farm gross income minus variable costs, both in Euro/farm), and farm labour demand (in working hours). These indicators were based on FSSIM modelling results and were determined by the combined effect of changes in crop yields, cropping patterns and prices, as described above. The input and product prices used were corrected for the inflation rate, being estimated at 1.9% per year over the period 2004 to 2050.

3.2.1. Farm net income

In Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalusia, climate change according to the *B1 scenario* results in, respectively, slight, slight, considerable and moderate increases and a strong decrease in farm net income. If the effects of technology and management improvements are also considered, CLIMT results in, respectively, strong, considerable, moderate, moderate and very strong increases in farm net income compared to those for CLIM (Fig. 6). The expected changes in prices (CLIMP) towards 2050 (Fig. 3) result in the five regions in, respectively, moderate, considerable, considerable, considerable and moderate increases in farm net income, compared to those for CLIM. CLIMTP with also technology improvement, resulting in lower product prices, results in the five regions in, respectively, slight, considerable, nil, slight and very strong increases in farm net income, compared to those for CLIM.

In the *A1-b1* and *B2* scenarios, the changes in farm net income compared to the base year are, respectively, essentially similar and moderately lower compared to those for the *B1* scenario in all regions except for Andalusia with its strongly variable results (Fig. 6). This is mainly due to the yield changes (Fig. 2) and price changes to 2050 (Fig. 3) that are approximately similar per crop product and per farming input for the *A1-b1* and *B1* scenarios and often moderately lower for the *B2* scenario. Farm net income in Andalusia for both base year and scenarios is almost nil which explains the strong differences in net income between the variants (Fig. 6).

Strong differences in net farm income between variants can, in addition to price and yield change effects, be explained to a large extent by changes in cropping patterns (Fig. 5 and Table 5).

3.2.2. Farm labour demand

In Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalusia climate change (CLIM) with the *B1 scenario* results in, respectively, moderate, nil, slight, nil and nil increases in farm labour demand. The increase in labour demand in Flevoland is due to the increasing area with seed potato at the expense of the areas of most other crops (Table 5). If the effects of technology and management improvements are also considered, CLIMT results in, respectively, moderate (due to further specialization in seed potato in Flevoland), moderate (due to specialization in grain maize in Midi Pyrenées), slight, nil and nil increases in farm labour demand compared to those for CLIM (Fig. 7). The expected changes in prices (CLIMP) towards 2050 result in the five regions in, respectively, a slight decrease (due to decreasing seed

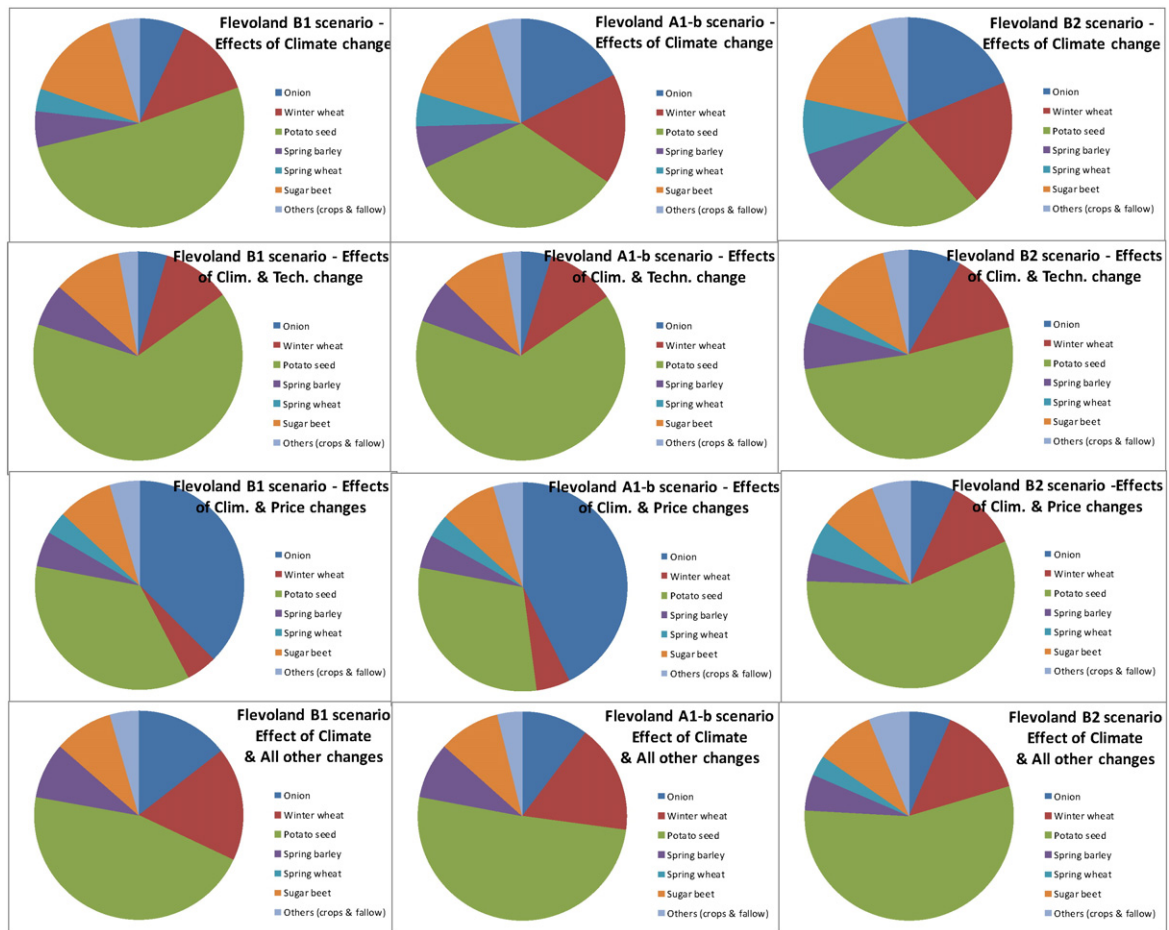
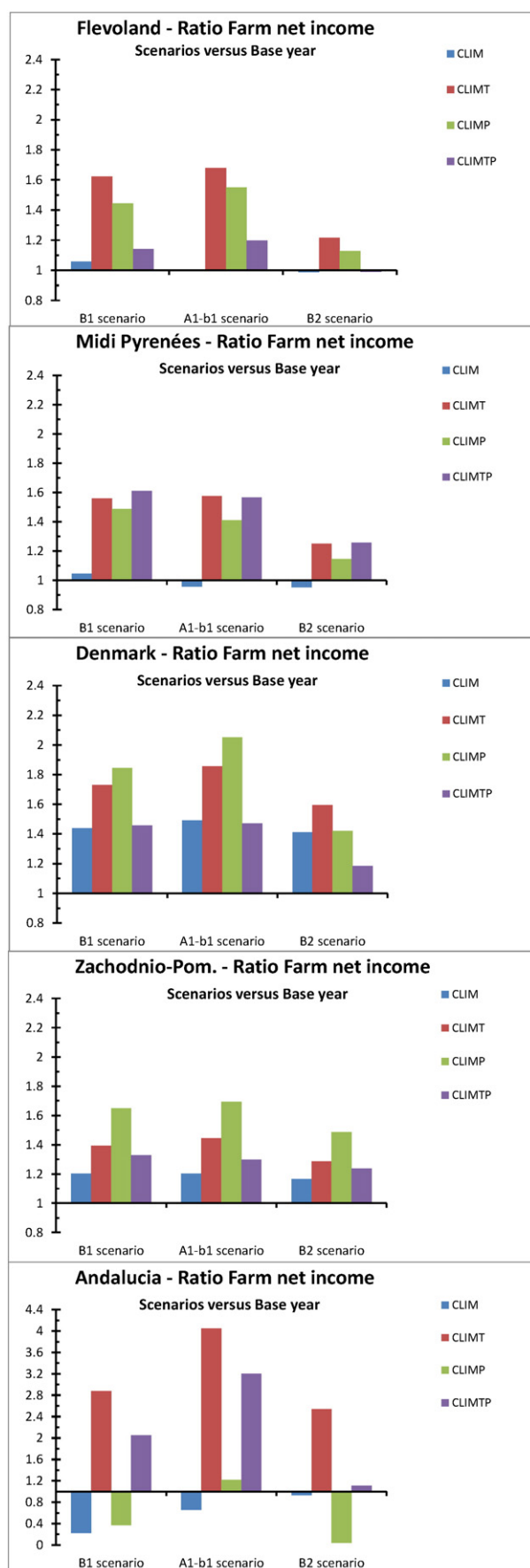


Fig. 5. Cropping patterns for typical farms in Flevoland for the baseline (i.e. B1), A1-b1 and B2 scenarios to 2050, considering the effects of (i) changes in climate and atmospheric CO₂ on crop yields only (CLIM; top figure), (ii) CLIM and changes in technology and management on crop yields (CLIMT; second figure), (iii) CLIM and changes in product and input prices (CLIMP; third figure) and (iv) CLIM and changes in all other factors (CLIMTP; bottom figure).



potato area in Flevoland, see Table 5), and moderate, slight, slight and moderate (due to increasing areas with oranges and olives in Andalucia) increases in farm labour demand, compared to those for CLIM. With technology improvement (CLIMTP), results for farm labour demand also show a slight decrease in Flevoland and moderate, slight, slight and moderate increases in, respectively, Midi Pyrenées, Denmark, Zachodniopomorski, and Andalucia compared to those for CLIM.

In the A1-b1 and B2 scenarios the changes in labour demands for 2050 in the five regions compared to those in the base year, are generally similar to those for the B1 scenario (Fig. 7). These results show that the changes in labour demand, as caused by the changes in cropping pattern, are mainly due to the differences between the four variants and much less so due to the differences between the three scenarios. Exceptions include the strong increase in seed potato area and thus in farm labour demand in Flevoland with climate change (CLIM) for the B1 scenario (Fig. 7) compared to the slight to moderate decreases in seed potato area (Fig. 5) and thus in farm labour demand for the A1-b1 and B2 scenarios, respectively. In addition, the increases in orange plantation areas and resulting labour demand in Andalucia with price changes towards 2050 (i.e. CLIMP and CLIMTP, Fig. 7) are moderate for the B1 and A1-b1 scenarios but not for the B2 scenario.

3.3. Changes in environmental indicators

The environmental indicators used in our analysis are ammonia (NH_3) and nitrous oxide (N_2O) emissions to the air and the sum of N leaching and N runoff to ground and surface water (all in $\text{kg N ha}^{-1} \text{yr}^{-1}$). N losses to air and water are largely determined by the total N inputs from applied fertilizers and animal manure. Changes in total N inputs appear to be mainly caused by the yield changes due to climate change of the main crops in each region (Fig. 2) as well as by changed cropping patterns to a less extent.

3.3.1. Nitrogen losses to the air

Changes in N_2O (see Appendix, Fig. A-4) and NH_3 (Fig. 8) emissions were determined from changes in both N_2O and NH_3 emission factors and the total N inputs from applied fertilizers and animal manure. We assumed that NH_3 emission fractions decreased to 2050 due to improvement of manure application techniques in Flevoland, Midi Pyrenées and Denmark. Hence, the relative changes in NH_3 emissions (Fig. 8) for the different scenarios and variants compared to those in the base year, showed very similar patterns to those for the N inputs in the five regions (see Appendix, Fig. A-3), though with smaller increases or larger decreases. In Andalucia and Zachodniopomorski, the relative changes in NH_3 emissions were essentially similar to those of the N inputs, as no improvements of manure application techniques were assumed and animal production and thus manure application were negligible in these regions (inorganic N fertilizers have a low NH_3 emission factor).

3.3.2. Nitrogen losses to water

Changes in the sum of N leaching and runoff to ground and surface water were determined by changes in both the N leaching and runoff factors (Table 4) and the N surplus (i.e. N inputs from applied fertilizers and animal manure minus the total crop N uptake). As with N_2O , the changes in the N leaching and runoff factors towards 2050 were limited, as they were mainly determined by relatively constant environmental factors, such as soil type, land use, soil organic content and rooting depth. These factors also depended on precipitation surplus and temperature, which changed over time, however, their assumed impact

Fig. 6. Relative changes in farm net income (inflation-corrected) on mean farms in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalucia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering the variants CLIM, CLIMT, CLIMP and CLIMTP (see Table 2).

was limited and not included in the current study. Consequently, the changes in N leaching to ground and surface water for the different scenarios and variants strongly followed the changes in N inputs (Fig. 9 versus Fig. A-3). However, due to the assumed increase in N use efficiency by 20% up to 2050, the increases in N leaching were slightly weaker and the decreases slightly stronger in most regions than the changes in N inputs. Only in Andalusia, the decreases in N leaching were weaker than the decreases in N inputs, because the relative contribution of natural N supply to N leaching became more important in 2050.

4. Discussion and main outcomes

4.1. Benefits of the integrated approach

Our study presents an integrated modelling approach to understand how European farming may look in 2050 considering climate change, price and technology changes. This approach goes beyond earlier integrated modelling approaches on climate change impacts in Europe. Its main benefits are first, assessment of climatic change impacts on agriculture in the context of technological, socio-economic and policy changes towards 2050 for regions over EU-27, second, impacts on the environment simulated with the INTEGRATOR model and not based on environmental indicator calculations and third, coherent linking of the four applied models, particularly with respect to crop yields and nitrogen use. This aspect is seen as particularly critical for integrated assessments, as for example each model is sensitive to estimate of nitrogen use. This has led to a strongly improved usability of the linked models for doing impact assessments for future conditions over Europe and for answering questions about the impacts of climatic, technological and economic changes on agriculture and the environment in the future.

4.2. Impacts of future changes in climate, technology and prices between EU regions

Results show that the effects of the three scenarios and the four variants on all socio-economic, agronomic and environmental indicators are different across the five regions. This has implications for policies which need to account for these regional differences in order to adequately anticipate future changes in climate, technology and prices. For example, climate change only under the B1 scenario results in considerable, slight and slight increases in farm net income in, respectively, Denmark, Flevoland and Midi Pyrenées, whereas strong increases are expected in all three regions, if improved technology is applied. When each of climate, technology and price changes is taken into account, considerable, slight and strong increases in farm net income result. Analyses of the model results also provide insight in the causes for these differences. Changes in farm income are caused by changes in crop yields, cropping pattern and prices for inputs and outputs. Shifts in farm labour demand are largely explained by changes in cropping pattern, which in turn are influenced by the different yield and price changes. Changes in N emissions and N leaching are mainly caused by changes in total N inputs from the applied fertilizers and animal manure, in turn determined by changes in N demand due to changes in crop yields and cropping patterns. Considering these different influencing variables, it is understandable that the modelled results are highly variable between scenarios, variants and regions.

4.3. Importance of considering technology development in farming systems

Model results show the importance of considering all the most relevant future changes in climate, technology, management and prices in scenario analyses, as done in the current study in combination and separately in the different variants. Particularly, it points to the relative importance of technology development already apparent from earlier studies (e.g. Ewert et al., 2005; Hermans et al., 2010; Angulo et al.,

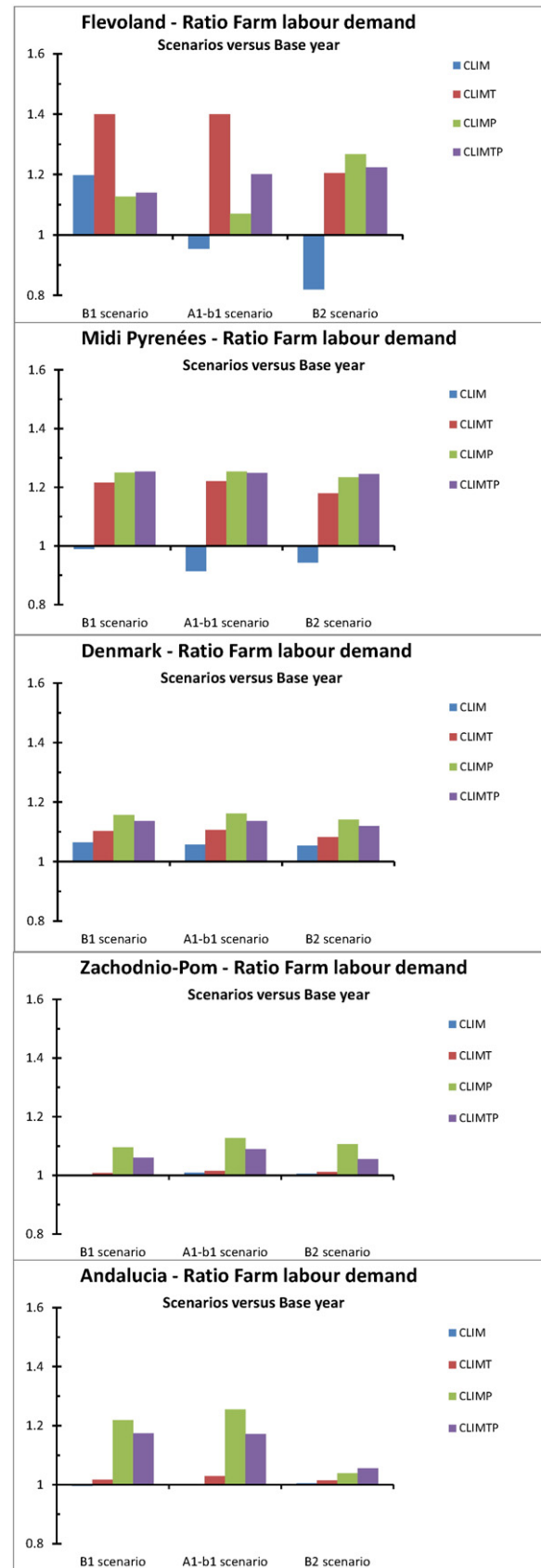


Fig. 7. Relative changes in farm labour demand on mean farms in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalusia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering the variants CLIM, CLIMT, CLIMP and CLIMTP (see Table 2).

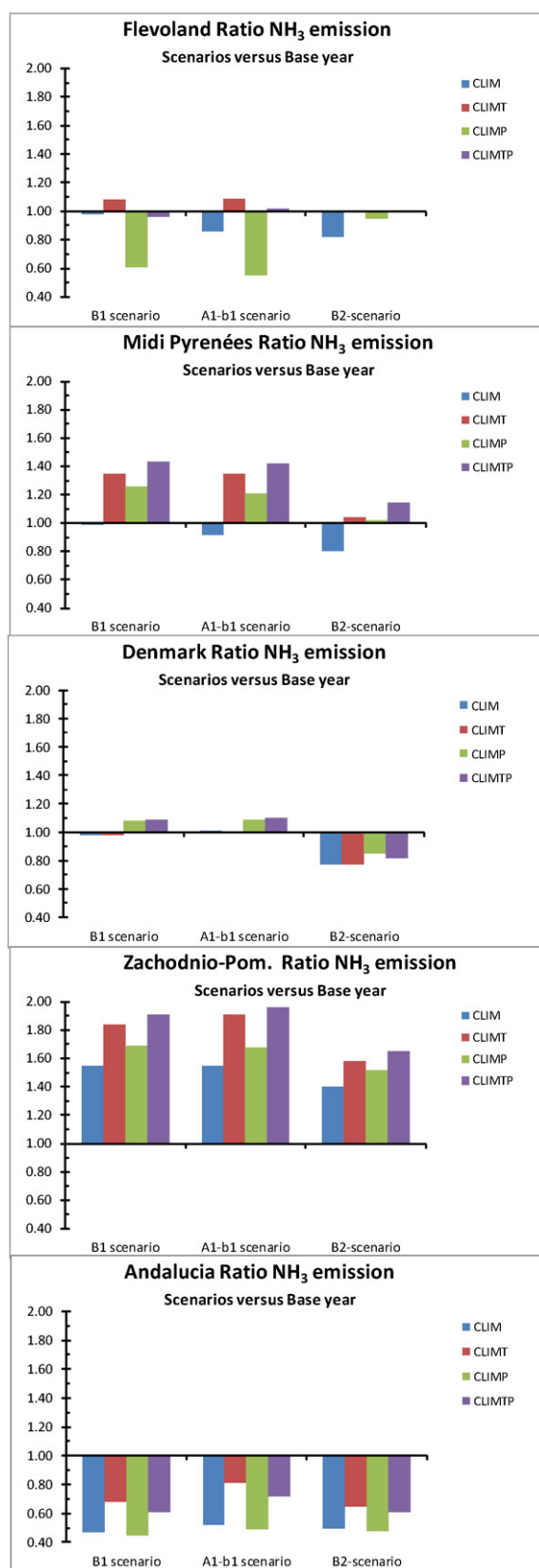


Fig. 8. Relative changes in ammonia emissions in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalucia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering the variants CLIM, CLIMT, CLIMP and CLIMTP (see Table 2).

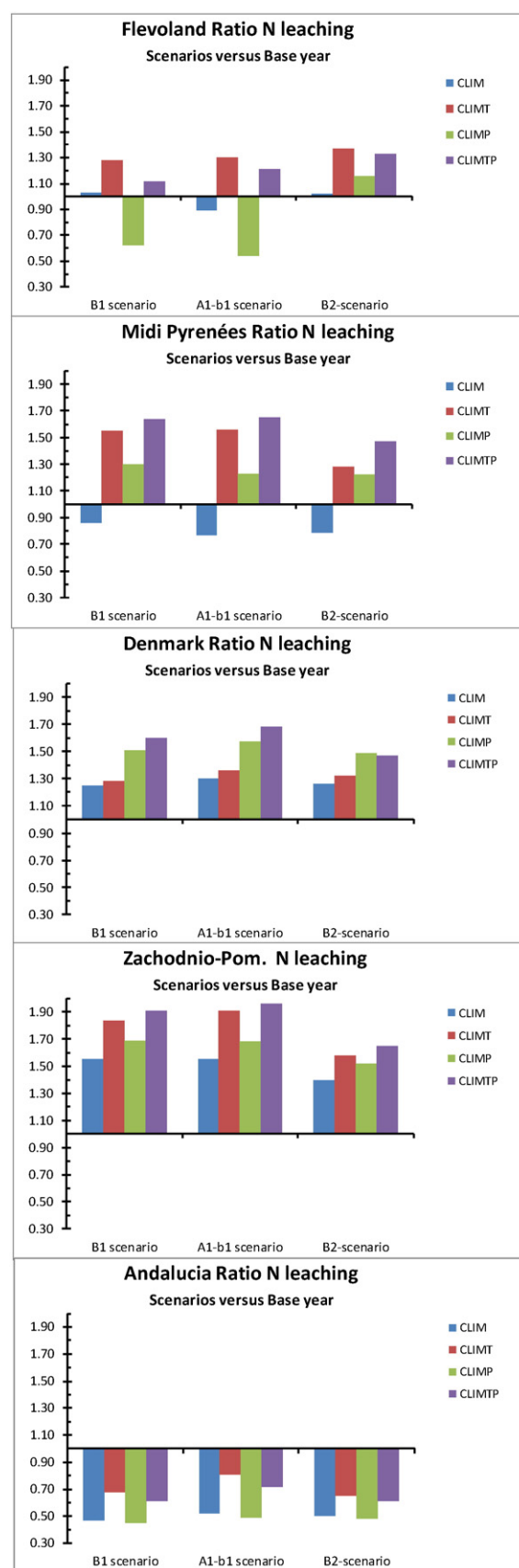


Fig. 9. Relative changes in the sum of N leaching and runoff to ground water and surface water in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalucia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering the variants CLIM, CLIMT, CLIMP and CLIMTP (see Table 2).

2013), and it stresses the need to reduce uncertainty in modelling and representing technology development in future scenarios. This can only be done by applying an integrated approach, as presented.

4.4. Policy implications of the modelling study

The differences in responses of a range of farming systems in the five regions over Europe point to the conclusion that policies should not manage markets or support specific farm branches, but rather foster the ability of farming systems to adapt to changing bio-physical and socio-economic conditions. This is best done by supporting innovation and technology adaptation, e.g. by support to life-long learning and fostering extension services and targeted research activities. At the same time, the CAP should set a clear and coherent framework to allow for regional and national targeted policies to reduce negative externalities (e.g. less environmental impacts) and support positive ones (e.g. increased farm incomes and sustainability). Such a policy framework should encourage technological development as a determinant factor for the future of EU farming systems. For example, new crop varieties that increase the potential yield levels and improved crop management practices that reduce the gap between potential and actual yields, increase the input use efficiencies and decrease the environmental emissions, can compensate for negative impacts of climate change (especially in southern EU) and can support farmer's adaptation strategies across the EU.

These conclusions are in line with existing more generic level conclusions summarized in e.g. the IPCC AR 5, "Approved summary for policy makers". The importance of technology development is one of the main conclusions in this summary, including the statement that "Transformations in economic, social, technological, and political decisions and actions can enhance adaptation and promote sustainable development (high confidence)." This study supports such conclusions and makes them more concrete in an agricultural context by performing integrated assessments based on application of well-validated models of different disciplines.

4.5. Methodological challenges

In conducting the integrated exercise, various conceptual and methodological challenges were encountered, particularly around the specification of future nitrogen use and consistent nitrogen indicators between models. A conceptual challenge was in the specification of how nitrogen use would evolve in the future. Future nitrogen rates are likely to be influenced by climate change impacts, but simulating climate change impacts requires first an estimate of nitrogen use. To circumvent this issue, our methodology included an adjustment of the future nitrogen rates by the same amount as the change in water limited yields, though this simplification neglects farmers' risk assessment processes in decision making, as well as, possible interactions between climate change and nitrogen use. Another, more methodological challenge, was in maintaining consistent values of nitrogen inputs between the models. For example, SIMPLACE simulates yields in response to climate and CO₂ changes, whereas FSSIM, CAPRI and INTEGRATOR use SIMPLACE yields with a historical yield trend superimposed (adjusted by scenario). Hence, different nitrogen inputs were needed for SIMPLACE which did not include the yield trends, than for the other models. While we were able to adjust nitrogen inputs for the different models, this issue precluded a more direct coupling and iteration between the models.

5. Conclusions

Integrated model assessments were carried out for three scenarios (i.e. baseline (B1) scenario, a strong economic growth A1-b1 and a weak economic growth B2 scenario) and considered the effects of climate change only (CLIM), climate change and technology and

management changes (CLIMT), climate change and price and policy changes (CLIMP), and climate, technology and price changes (CLIMTP). The assessments gave the following main results:

a. Farm income

- Climate change towards 2050 results in higher farm net incomes in the Northern and Northern-Central EU regions (e.g. Denmark and Zachodniopomorski), in practically unchanged farm net incomes in the Central and Central-Southern EU regions (e.g. Flevoland and Midi-Pyrénées) and in much lower farm net incomes in Southern EU regions (e.g. Andalucia).
- Climate change in combination with improved technology and farm management towards 2050 results in much higher farm net incomes in all five EU regions compared to those in the base year.
- Climate change in combination with price changes towards 2050 results in much higher farm net incomes for the B1 and A1-b1 scenarios and in higher farm net incomes for the B2 scenario in four out of five EU regions (with Andalucia as exception) compared to those in the base year.
- With the all including CLIMTP variant the farm net incomes in the five regions moderately to strongly increase for the B1 and A1-b1 scenarios and zero to moderately increase for the B2 scenario compared to those in the base year.
- Increases in farm net income for the B1 and A1-b1 scenarios are moderately stronger than those for the weak economic growth B2 scenario; these differences are caused by the smaller increases in product prices and/or yields for the B2 scenario.
- Changes in farm income towards 2050 are generally affected by the changes in the following factors, which are given in the order of that with the largest to that with the smallest impact: product prices, cropping pattern, technology and farm management, climate.

b. Farm labour demand

- Farm labour demand generally increases towards 2050 as related to the changes in cropping pattern, which depend on the scenario, variant and region. Labour demands moderately increase in Flevoland and Midi-Pyrénées compared to those in the base year due to their specialization in, respectively, seed potato plus onion and grain maize and slightly increase in Denmark and Andalucia due to their specialization in, respectively, winter wheat plus grass and in orange plantations for only the A1-b1 and B1 scenarios.
- Differences in the changes in farm labour demand towards 2050 are mainly due to the differences between the four variants and much less so due to the differences between the three scenarios.

c. N losses to air and water

- Changes in N₂O emissions and N leaching towards 2050 are mainly caused by the changes in total N inputs from the applied fertilizers and animal manure, which in turn are influenced by changes in crop yields and cropping patterns, whereas NH₃ emissions are mainly determined by assumed improvements in manure application techniques.
- N emissions and N leaching appear to increase or decrease towards 2050 depending on the scenario, variant and region. In general, they strongly increase in Denmark and Zachodniopomorski, slightly decrease to moderately increase in Flevoland and Midi-Pyrénées, and strongly decrease in Andalucia, except for NH₃ emissions which nil to moderately decrease in Flevoland and Denmark.

d. Role of EC policies

- Impacts of EC policies on EU farming systems are expected to become more and more limited over time. This is due to the CAP reforms which has resulted in a complete decoupling of the farm payments, expiry of the quota, etc.
- Future developments in EU farming will be mainly driven by the possibilities to increase farm income and to decrease labour demand. In arable farming systems labour demands are rather limited. Hence, EC policies that are meant to, for example, increase sustainability of

arable farming systems and decrease their emissions, should focus on the main factors influencing farm income. These factors on which EC policies should focus, are given in the order of that with the largest to that with the smallest impact: product prices, cropping pattern, technology and farm management. However, given the continuing market integration and liberalization, a focus on product prices, i.e. direct market intervention, hardly makes sense in practice. Hence, EC policies should support farmers to adjust to future markets and other developments by fostering innovation and technology adaptation such that farmers successfully can adjust cropping patterns and more general their farm management.

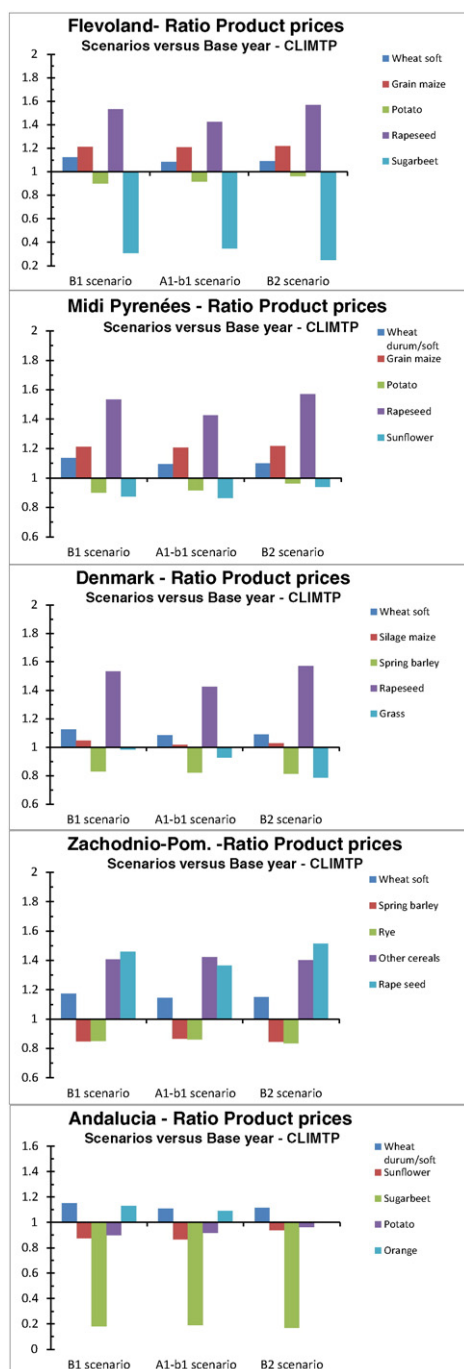


Fig. A-1. Relative changes in inflation-corrected (assuming inflation rate of 1.9% per year) prices of the main crop products on mean farms in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalucia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005) as computed with the CAPRI model – technology change & variant CLIMTP.

Integrated model assessments of future scenarios are shown to provide useful information which can be used to assess the consequences of future changes at multiple scale levels (from EU to region to field). Of course, the interpretation of such results should be linked to specific assumptions and scenario settings. The differences in responses of a range of farming systems in the five regions over Europe to future changes (as it was shown in this study) and the differences in possible future scenarios, e.g. with regard to product prices, point to the conclusion that policies should not manage markets or support specific farm branches, but

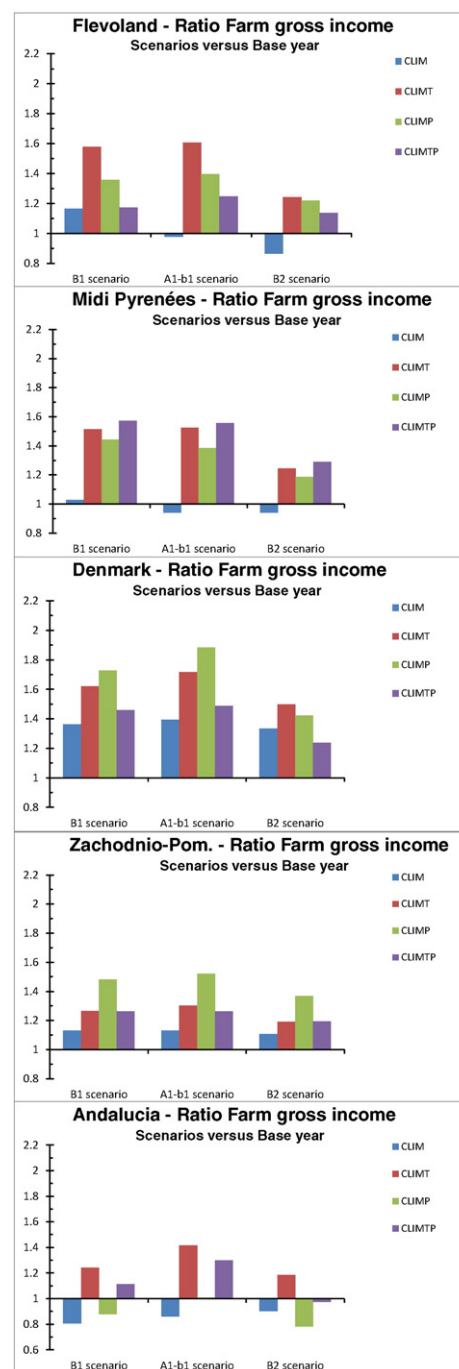


Fig. A-2. Relative changes in farm gross income (inflation-corrected) on mean farms in Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalucia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering effects of (i) climate change and increased atmospheric CO₂ only (CLIM), (ii) both CLIM and technology and management changes (CLIMT), (iii) both CLIM and changes in prices (CLIMP), and (iv) CLIM and all changes in technology/management and prices (CLIMTP).

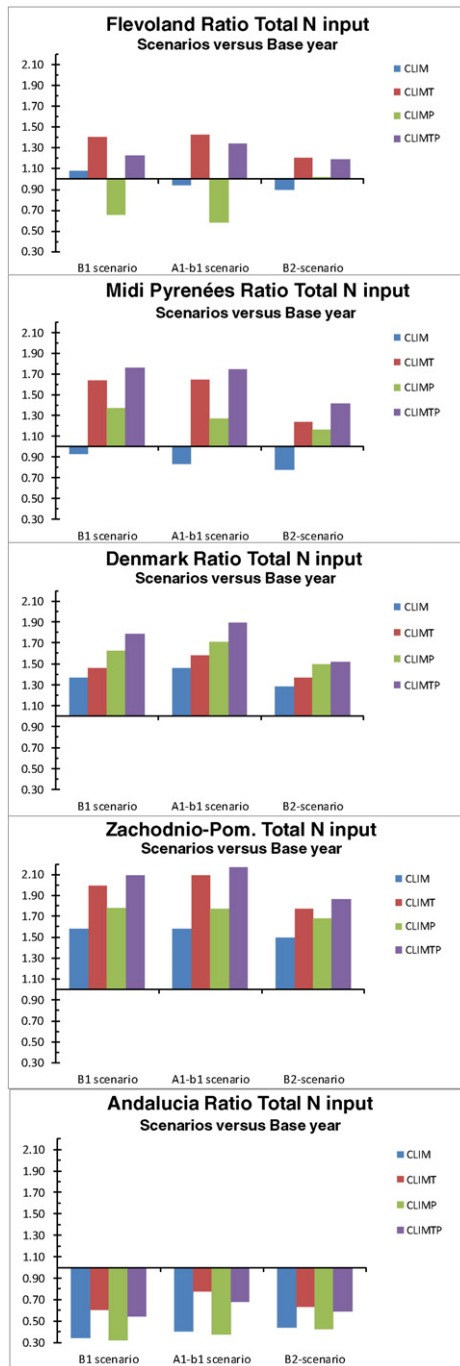


Fig. A-3. Relative changes in total N input in Flevoland, Midi Pyrénées, Denmark, Zachodniopomorski and Andalusia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering the variants CLIM, CLIMT, CLIMP and CLIMTP (see Table 2).

rather foster the ability of farming systems to adapt to changing biophysical and socio-economic conditions.

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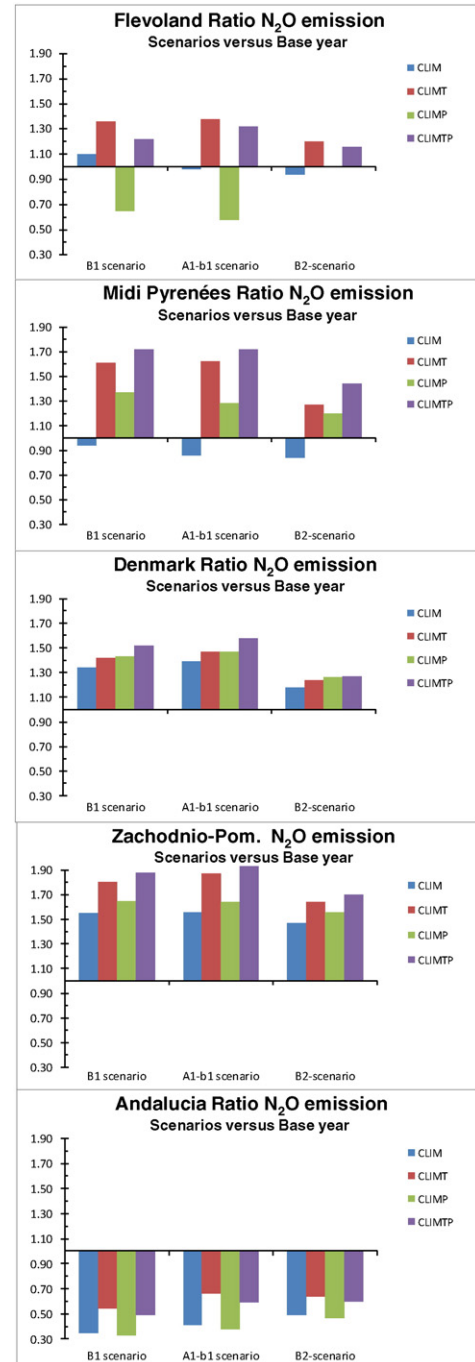


Fig. A-4. Relative changes in nitrous oxide emissions in Flevoland, Midi Pyrénées, Denmark, Zachodniopomorski and Andalusia for, respectively, the baseline (i.e. B1), A1-b1 and B2 scenarios for 2050 compared to the base year (i.e. 2003–2005), considering the variants CLIM, CLIMT, CLIMP and CLIMTP (see Table 2).

Appendix A

Additional information is given here about (a) prices of agricultural products for CLIMTP variant, (b) farm gross income, (c) nitrogen inputs and (d) N₂ losses to the air.

A.1. Prices of agricultural products for CLIMTP variant

Price changes are derived from CAPRI simulations for a situation in which farm technology (e.g. improved crop varieties) and management improvements are assumed (Fig. A-1; variant CLIMTP). The product prices appear mainly to change by a factor between 0.9 and 1.5 from 2004 to 2050. These smaller price increases compared to those for variant CLIMP (Fig. 3) are due to the stronger increases in crop production with farm technology and management improvements (Fig. 2). The differences in product price changes between the different crops and regions for the CLIMTP variant (Fig. A-1) are similar to those described above for the CLIMP variant (Fig. 3).

A.2. Farm gross income

In Flevoland, Midi Pyrenées, Denmark, Zachodniopomorski and Andalusia climate change (CLIM) for the B1 scenario results in slight, nil, moderate and slight increases and a moderate decrease in farm gross income, respectively. If the effects of technology and management improvements are also considered, CLIMT results in, respectively, considerable, considerable, moderate, slight and considerable increases in farm gross income compared to those for CLIM (Fig. A-2). The expected changes in inflation corrected prices in 2050 (CLIMP) result in the five regions in, respectively, moderate, considerable, moderate, moderate and slight increases in farm gross income, compared to those for CLIM. CLIMTP which includes technology improvement and the resulting lower product prices (Fig. A-1 versus Fig. 3), results in the five regions in, respectively, nil, considerable, slight, slight and moderate increases in farm gross income, compared to those for CLIM.

In the A1-b1 and B2 scenarios the changes in farm gross income compared to the base year are respectively, essentially similar and moderately lower compared to those for the B1 scenario in all regions (Fig. A-2). This is mainly due to the yield changes (Fig. 2) and price changes to 2050 (Fig. 3) that are similar per crop for the A1-b1 and B1 scenarios and often moderately lower (but not in case of price changes for CLIMTP in Fig. A-1) for the B2 scenario.

A.3. Nitrogen inputs

With respect to the total N inputs slight to moderate decreases in response to climate change were modelled for all three scenarios in Midi Pyrenées and zero to slight decreases for the scenarios in Flevoland (Fig. A-3). Strong decreases in total N input for all three scenarios were simulated for Andalusia, whereas for Denmark and Zachodniopomorski strong increases were modelled. These changes in total N inputs were mainly caused by the changes in N inputs in response to the yield changes due to climate change of the main crops in each region (Fig. 2) as well as by changed cropping patterns to a less extent. When also the effects of improved technology and management (CLIMT) in 2050 were considered, moderate to strong increases in N inputs were found for all regions and scenarios, compared to the N inputs in the base year, except for Andalusia where results indicate a strong decrease. This can be explained too from mainly the yield changes of the main crops by climate and technology change (Fig. 2) and the adapted N inputs. Price changes to 2050 (i.e. CLIMTP vs. CLIMT and CLIMP vs. CLIM) in Flevoland led to zero to moderate decreases in applied N inputs, to zero changes in Andalusia, and in Midi Pyrenées, Denmark and Zachodniopomorski to moderate to considerable increases in applied N inputs. These changes in total N inputs due to the price changes were mainly due to the modelled changes in cropping

patterns, such as the decreasing seed potato area in Flevoland resulting in lower N inputs, slight changes in cropping pattern in Andalusia, and specialization in grain maize in Midi Pyrenées, in grass fodder in Denmark and in rape seed in Zachodniopomorski (see Table 5), all of which resulted in higher N inputs.

A.4. N₂ losses to the air

Changes in N₂O emissions (Fig. A-4) were determined from changes in both the N₂O emission factor and the total N inputs from applied fertilizers and animal manure. The N₂O emission fractions hardly changed by 2050, as they are mainly determined by relatively constant environmental factors, such as N source, soil type, land cover and precipitation. Note that we did not adjust precipitation in the INTEGRATOR modelling for the three scenarios. Consequently, the changes in N₂O emissions for the different scenarios and variants strongly followed the changes in N inputs (Fig. A-4 versus Fig. A-3).

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